

Experiment - 7 16-Bit Multiplication.

AIM: To write an assembly language program to implement 16-bit multiplication using 8085 processor.

ALGORITHM:

1. load the first data in HL pair.
2. move content of HL pair to stack pointer
3. load the second data in HL pair and move it to DE.
4. make H register as 00H and L register as 00H.
5. Add HL pair and stack pointer.
6. check for carry if carry increment it by 1 else move to next step.
7. Then move E to A and perform OR operation with accumulator and register D.
8. The value of operation is zero, then store the value else go to step.

PROGRAM:

LHLD 2050 - loads the contents of memory location 2050H and 2051H into the HL register pair.

SPHL - sets the stack pointer (SP) to the value in the HL register pair.

LHLD 2052 - load second 16-bit data.

XCHG - Exchange HL and DE

LXI H, 0000H - clear HL register pair

LXI B, 0000H - clear BC register pair

AGAIN: DAD SP - Add SP to HL

JNC START - Check carry flag.

INX B - Increment BC register.

START: DEX D - Decrement DE register.

MOV A, E - move E into accumulator

ORA D - OR A with D.

JNZ AGAIN - Repeat if non zero

SHLD 2054 - store result in memory

MOV L, C - move C into L.

MOV H, B - move B into H.

SHLD 2055 - store carry in memory.
HLT - Stop program execution.

INPUT:

Address	Data
2050	10
2052	5

OUTPUT:

Address	Data
2054	50
2055	5

Result: Thus the program was executed successfully using 8085 processor simulator.

FileResetAssemblerDebugHelp

Registers

A00

BC0000

DE0000

HL0000

PSW0000

PC4222

SP000A

Int-Reg00

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0-+00

Update Port Value

Memory

2052-+05

Update Memory

Load me at

1LHLD 2050

2SPHL

3LHLD 2052

4XCHG

5LXI H,0000H

6LXI B,0000H

7AGAIN: DAD SP

8JNC START

9INX B

10START: DCX D

11MOV A,E

12ORA D

13JNZ AGAIN

14SHLD 2054

15MOV L,C

16MOV H,B

17SHLD 2055

18HLT

DataStackKeyPadMemoryI/O Ports

Start2054OK

Address (Hex)	Address	Data
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0
0814	2068	0
0815	2069	0

Line NoAssembler Message

0Program assembled successfully